

Epoch Ledger

Every completed epoch so far, stage 1 through stage 2 (in progress) - loss, ordinal agreement, minority-class recall, screening/STDR discrimination, hard-example-mining state, and the full per-grade recall spread, read directly from the training log.

EPOCHS COMPLETE 16 / 45 Stage 2, epoch 12 of 12 in progress	BEST QWK 0.554 Stage 2, epochs 5 & 6	BEST SCORE 0.471 Stage 2, epoch 5	BEST MIN-CLASS RECALL 0.474 Stage 2, epoch 7	BEST REFAUC 0.848 Stage 2, epoch 9 - climbing steadily
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Column glossary

Time context Wall-clock minutes to train the epoch; not a quality signal.	Score higher better Composite selection score used to pick the best checkpoint.	Screen thr context Decision threshold that hit the NPV target this epoch.
Loss lower better Weighted sum of every training objective, averaged per image.	Accuracy higher, cautiously Overall correct-grade rate; can rise just by favouring grade 0.	STDR AUC higher better AUC for detecting sight-threatening DR (Severe + PDR).
QWK higher better Quadratic Weighted Kappa: ordinal agreement, penalises distant misses harder.	refAUC higher better AUC for the binary referable-vs-not-referable DR split.	Miner p90 H context 90th-percentile mining hardness carried into the next epoch.
F1 higher better Macro F1 across all 5 grades, treating each equally.	Screen cleared context Share of images passed without flagging; trades off against NPV.	Boundary rate context Share of samples flagged as one-grade boundary errors.
Min-recall higher better Recall of whichever grade is doing worst that epoch - the key imbalance metric.	Screen NPV higher better Negative predictive value of the screening gate vs a 0.985 target.	Per-grade recall higher better, all 5 Recall per grade 0-4 - see the dedicated heatmap page.

Full comparison, all 16 epochs

Stage 1: head only, backbone frozen. Stage 2: GLA-LoRA + ALPP trainable, backbone still frozen. Shading: worst (red) to best (green) per column; loss inverted (lower is greener).

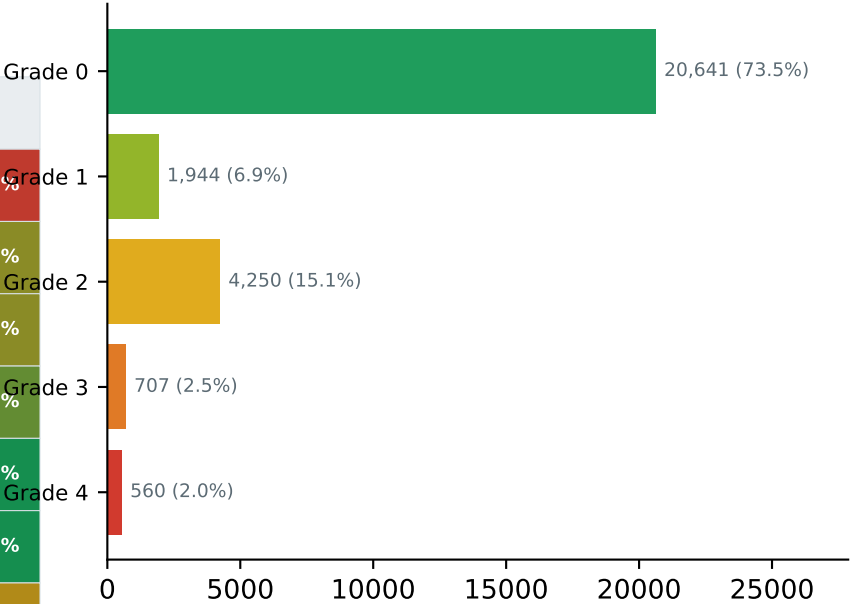
Epoch	Time	Loss	QWK	F1	Min-recall	Score	Accuracy	refAUC	Screen cleared	Screen NPV	Screen thr	STDR AUC	Miner p90 H	Boundary rate
S1 e1	75.7m	83.90	0.351	0.292	0.187	0.300	0.666	0.726	91.1%	0.9854	0.237	0.8877	1.00	0.03
S1 e2	68.8m	48.43	0.421	0.287	0.198	0.336	0.607	0.778	94.1%	0.9851	0.599	0.9162	1.00	0.05
S1 e3	75.6m	33.65	0.441	0.284	0.168	0.339	0.634	0.794	93.9%	0.9851	0.590	0.9165	1.15	0.06
S1 e4	75.6m	27.29	0.403	0.301	0.289	0.350	0.569	0.800	94.1%	0.9851	0.668	0.9149	1.28	0.07
S1 e5	70.1m	25.51	0.458	0.309	0.211	0.364	0.639	0.808	93.9%	0.9851	0.646	0.9176	1.37	0.08
S2 e1	83.4m	24.05	0.434	0.287	0.228	0.349	0.539	0.809	94.9%	0.9852	0.697	0.9203	1.43	0.08
S2 e2	95.0m	17.47	0.496	0.376	0.269	0.414	0.725	0.804	94.7%	0.9852	0.470	0.9163	1.47	0.08
S2 e3	85.7m	12.15	0.465	0.303	0.238	0.371	0.609	0.813	94.6%	0.9852	0.652	0.9201	1.52	0.08
S2 e4	98.8m	9.34	0.456	0.318	0.301	0.384	0.587	0.820	94.5%	0.9852	0.686	0.9215	1.55	0.09
S2 e5	100.8m	8.13	0.554	0.399	0.370	0.471	0.703	0.832	94.2%	0.9851	0.564	0.9461	1.58	0.09
S2 e6	91.5m	7.34	0.554	0.390	0.334	0.461	0.691	0.836	94.2%	0.9851	0.549	0.9345	1.59	0.09
S2 e7	98.4m	6.95	0.402	0.344	0.474	0.399	0.479	0.836	94.9%	0.9853	0.662	0.9459	1.61	0.09
S2 e8	91.0m	6.62	0.509	0.380	0.385	0.445	0.611	0.839	95.5%	0.9853	0.643	0.9413	1.61	0.09
S2 e9	96.6m	6.37	0.474	0.366	0.418	0.430	0.578	0.848	95.0%	0.9853	0.693	0.9426	1.60	0.09
S2 e10	83.5m	6.20	0.482	0.380	0.438	0.443	0.587	0.839	94.9%	0.9852	0.691	0.9410	1.60	0.09
S2 e11	92.2m	6.08	0.510	0.391	0.435	0.459	0.615	0.839	94.7%	0.9852	0.665	0.9499	1.61	0.09

Per-grade recall, every epoch

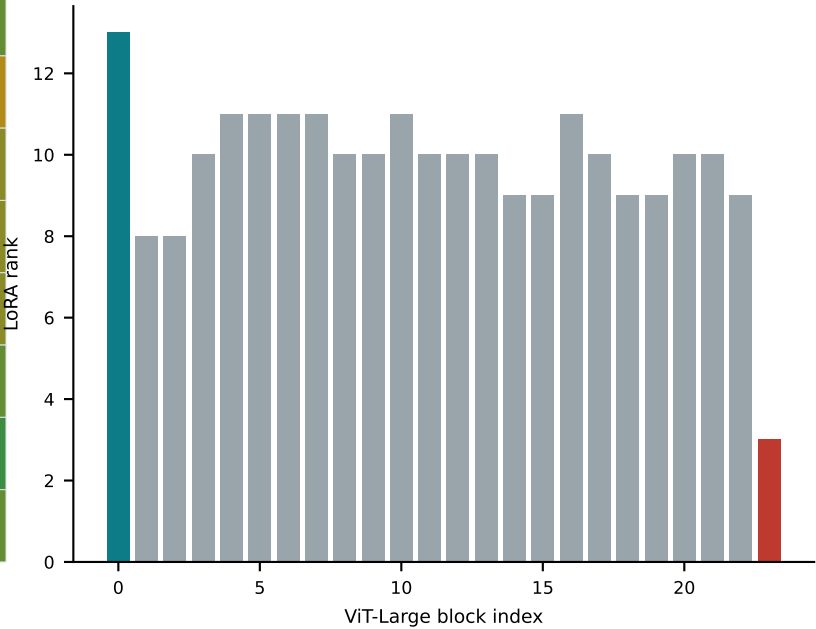
Shaded per grade (each column against its own best/worst across all 16 epochs) so a hard grade like Mild still shows its own real progress.

Epoch	Grade 0 No DR	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 PDR
S1 e1	82.9%	5.9%	22.4%	27.8%	0.0%
S1 e2	72.6%	5.9%	34.1%	19.4%	28.6%
S1 e3	76.7%	6.9%	32.2%	11.1%	28.6%
S1 e4	65.3%	10.9%	39.7%	36.1%	33.3%
S1 e5	75.8%	5.9%	37.9%	19.4%	42.9%
S2 e1	61.1%	14.9%	42.5%	11.1%	42.9%
S2 e2	89.5%	5.0%	22.9%	52.8%	23.8%
S2 e3	71.9%	9.9%	36.4%	25.0%	28.6%
S2 e4	67.6%	10.9%	40.7%	38.9%	33.3%
S2 e5	83.5%	5.0%	36.4%	69.4%	23.8%
S2 e6	82.4%	5.0%	34.1%	61.1%	28.6%
S2 e7	48.9%	21.8%	53.7%	66.7%	28.6%
S2 e8	69.5%	9.9%	44.4%	61.1%	28.6%
S2 e9	63.9%	13.9%	47.7%	63.9%	33.3%
S2 e10	65.6%	12.9%	43.5%	75.0%	38.1%
S2 e11	69.5%	11.9%	43.5%	75.0%	33.3%

Training-split grade distribution (28,102 images)

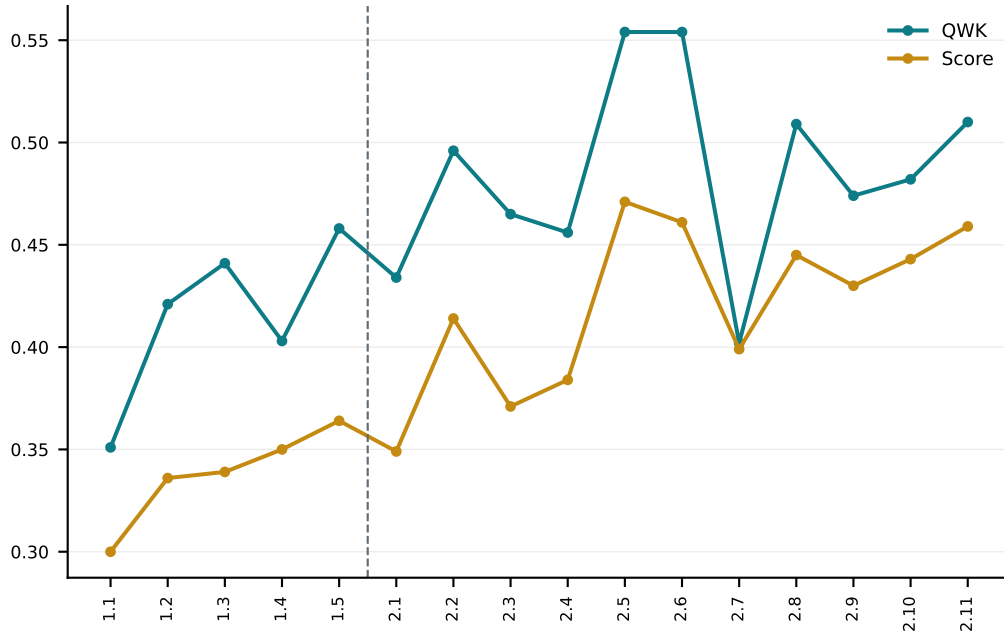


GLA-LoRA per-block rank allocation (real, gla_lora.json)

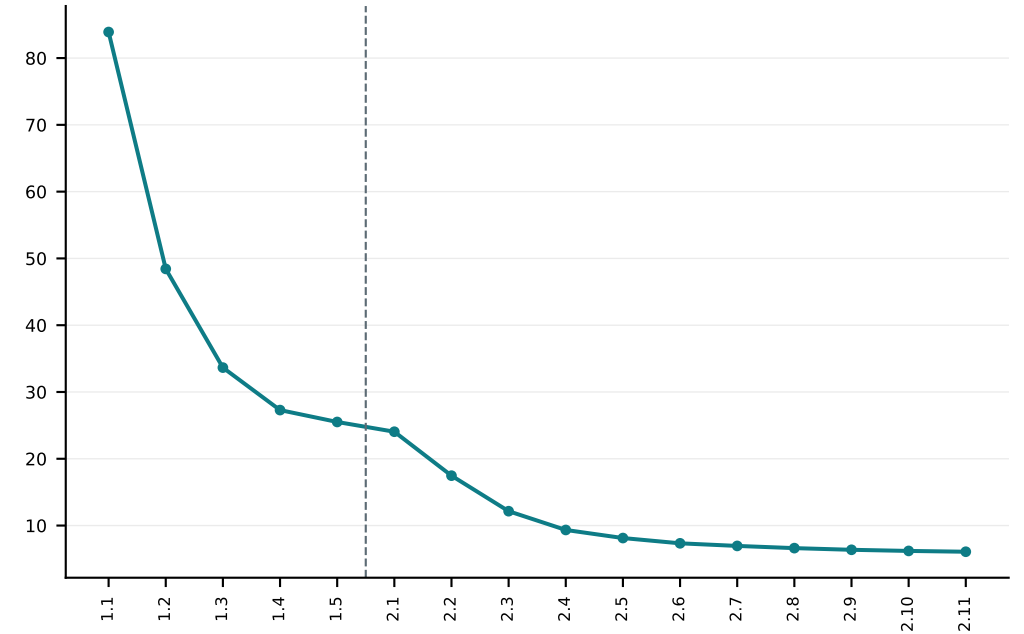


Trends

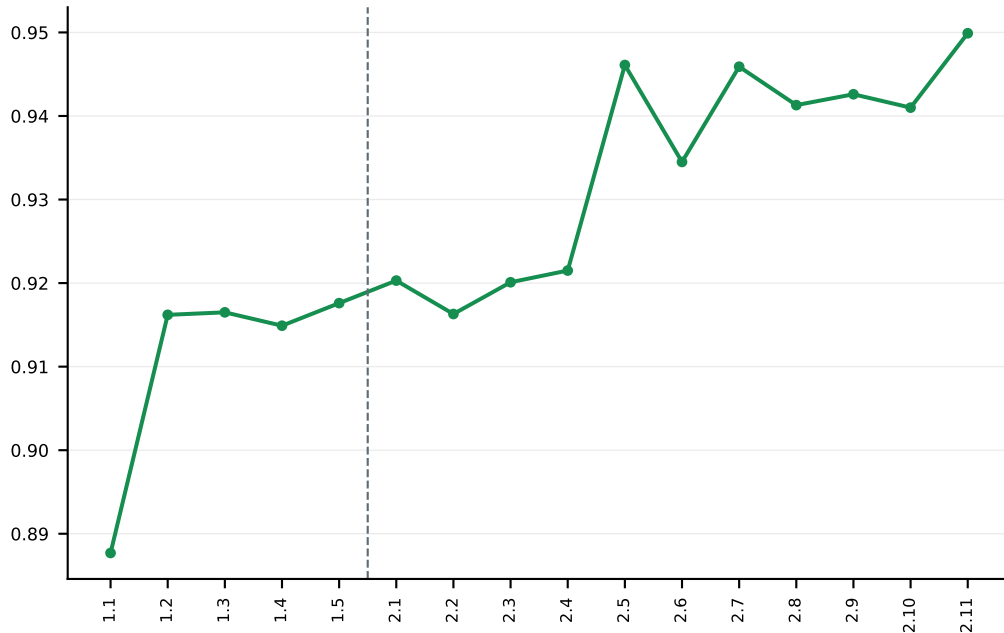
QWK & Score, all 16 epochs



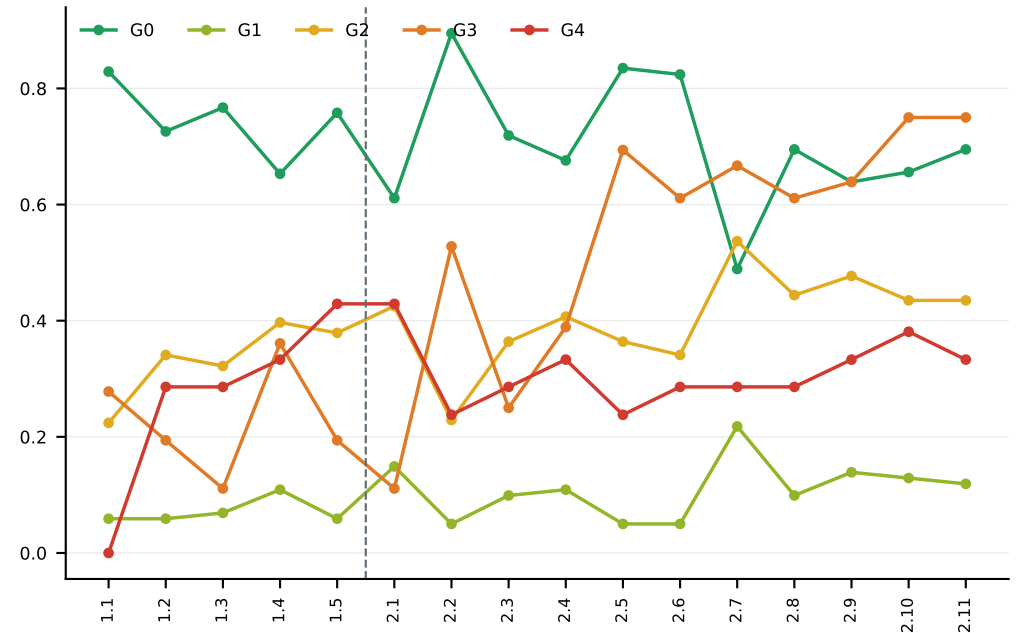
Loss, all 16 epochs



STDR AUC (sight-threatening DR), all 16 epochs



Per-grade recall trend, all 16 epochs



Notable events

Grade 1 (Mild) breaking its floor
Stuck at 5.11% recall for 12 straight epochs. Mild reached 21.8% (e7), 13.9% (e9), 12.9% (e10), and 11.9% (e11) - four of the last five epochs above the old ceiling. Too early to call it settled, but the first sign this isn't just noise.

Epoch 10's five failed attempts
Restarts kept landing before the ~90-100 minute epoch could finish and checkpoint, wiping it out each time across 4 days. Fixed by adding a checkpoint every 500 steps mid-epoch, plus (later) exact-step resume via a seeded sampler, instead of redoing the whole epoch.

Stage 1 -> Stage 2 lift
Turning on GLA-LoRA and ALPP training lifted every average metric: QWK 0.415->0.483, score 0.338->0.412, min-recall 0.211->0.331 - at the cost of ~27% longer epochs.

Setup & architecture reference

Dataset	35,126 images, 17,563 patients (EyePACS)
Split	28,102 / 3,512 / 3,512 train/val/test, no patient leakage
RETFound coverage	100% - 294/294 backbone tensors loaded
GLA-LoRA params	1,431,552 - 0.45% of the 319.6M backbone
Sampling ratio	5.01x grade4:grade0 weight, $n^{-0.5}$